

Homework III: Foundational Image Operations

Objective: Implement global image thresholding with morphological cleanup, and image downsampling via bilinear interpolation from scratch using MATLAB. Compare your custom algorithms against MATLAB's built-in functions.

General Requirement for ALL Parts: For every task in this assignment, you must:

1. Write your own algorithm from scratch using basic matrix operations and loops (do not use built-in MATLAB functions for the core logic).
2. Use built-in MATLAB functions to perform the exact same task.
3. Display the original image alongside the outputs of both your custom algorithm and the built-in function side-by-side.

Part 1: Segmentation and Morphological Cleanup

Setup: Load a grayscale image and synthetically introduce a small amount of "salt and pepper" noise.

Task: Implement Otsu's thresholding method from scratch to find the optimal global threshold (T) by maximizing inter-class variance, and generate a binary mask. Then, define a 3x3 square structuring element and write a custom algorithm to perform a Morphological Closing operation (Dilation followed by Erosion) on your mask to reduce noise and connect regions.

Compare: Perform the equivalent segmentation and morphological cleanup using MATLAB's built-in functions (e.g., `graythresh`, `imbinarize`, `imclose`, `strel`).

Analysis: Explain the mathematical reasoning behind the morphological closing step. Discuss why the specific sequence of operations (Dilation then Erosion) is effective for connecting objects versus removing background noise.

Part 2: Image Compression (Bilinear Downsampling)

Setup: Load a high-resolution image and convert it to double precision.

Task: Implement a custom image downsampling algorithm using Bilinear Interpolation to reduce the spatial dimensions of the image by exactly 50%. Your algorithm must compute new pixel intensities using the weighted average of the four nearest spatial neighbors and appropriately handle edge boundary conditions.

Compare: Use MATLAB's built-in `imresize` function (with the method explicitly set to 'bilinear') to achieve the same 50% scale reduction.

Analysis: Discuss the mathematical distinction between Nearest-Neighbor and Bilinear interpolation. Explain why Bilinear interpolation inherently acts as a low-pass filter and how that impacts the visual sharpness of the compressed image.

What to Submit

- Separate MATLAB scripts (.m) for Parts 1 and 2. Ensure your custom functions are included in the scripts or the same folder.
- The image files used in your code.
- A short PDF containing your written analysis for both parts.
- Please compress all files into a single .zip folder.